

| # | Milestone                   | Generative AI Guidelines            | Due                                             | Mark Release  |
|---|-----------------------------|-------------------------------------|-------------------------------------------------|---------------|
| 1 | Background and Related Work | C: AI-Assisted Editing <sup>1</sup> | Oct. 07, Wed., <a href="#">AoE</a> <sup>2</sup> | Oct. 12, Mon. |

This course is *project driven*. Students will work in teams to define, develop, and complete a project in the field of data mining/machine learning over the course of one semester. Each team will consist of undergraduate students and one graduate student, who serves as the project lead, coordinating the project and guide the team through the research and development process. The course project has two main objectives:

- Hands-on experience: Gain practical experience in applying data mining and machine learning methods to a real-world problem, while developing skills in project planning, implementation, evaluation, and teamwork.
- Research experience: Develop an understanding of the research *process*, from identifying a problem and reviewing existing approaches to designing, implementing, and evaluating a solution.

Projects should aim to advance the state of the art, rather than simply reproduce an existing solution. This provides students with an opportunity to conduct meaningful research while completing the course requirements.

A research project typically starts with a problem and proposes a solution by designing, implementing, and evaluating an algorithm or computational approach. In this sense, it may resemble a software project; however, there are important differences. A software project may primarily focus on implementing or integrating existing algorithms to build a functional system. In contrast, a research project should aim to investigate a problem, identify a limitation or gap in existing approaches, and propose a solution that addresses that gap. The proposed solution may involve (1) a new algorithm, (2) a modification or extension of an existing algorithm, (3) a new combination/pipeline of existing techniques, or (4) a novel application of existing methods to a problem where they have not been adequately explored. Therefore, the goal is not simply to implement an existing method, but to make a meaningful contribution beyond what has already been done, supported by appropriate experiments and evaluation. Roughly, a research project consists of the following milestones (phases):

1. Project Definition & Team Formation (Proposal)
2. Background & Related Work (Literature Review)
3. Solution Design & Implementation (Proposed Method)
4. Evaluation (Experimentation)
5. Project Presentation: 3-Minute Talk (3MT)
6. Project Management & Documentation (Graduate Student Only, Umbrella Activity)

A project manual has been prepared to guide students through each milestone. Each milestone breaks the project into a series of smaller steps and deliverables. The current manual covers Milestone 2: Background & Related Work, also known as Literature Review.

<sup>1</sup> AI can be used to make improvements to the clarity or quality of student-created work to improve the final output, but no new content can be created using AI (e.g. language editing for lab reports). AI can be used, but your original work with no AI content must be provided in an appendix.

<sup>2</sup> <https://www.timeanddate.com/time/zones/aoc>

### 1.1. Introduction

The purpose of this milestone is to review what has already been done on your project topic and to place your project in the context of existing work. You should explain:

- What approaches have been used to address the problem in the project?
- How are these approaches different?
- What are their strengths and limitations?
- Where does your project fit among these existing approaches?

The review is a *map* of the existing solution space, rather than a collection of summaries on existing works.

### 1.2. History

You need to include a brief historical review of the origin and evolution of the problem and provide context for a reader who is not familiar with the topic. A short historical perspective can help with where the problem came from, how it evolved, and why it eventually became an important problem for computer science and data mining/machine learning.

You should trace the problem back to its earliest known roots, even if these roots predate computer systems. Look for the earliest ideas, practices, or challenges that are related to the problem you are studying. You do *not* need to provide an exhaustive history; the goal is to give the reader a quick journey (odyssey) from the early origins of the problem to its modern computational form. For example, the history of numerical representation can be briefly described as:

*“Early numbering systems were developed by ancient civilizations in 1<sup>st</sup> and 4<sup>th</sup> centuries. The decimal system was subsequently developed and refined through Indian and Arabic mathematics before evolving into the numerical system widely used today.”*

Similarly, if your project concerns a modern machine learning problem, you might begin with the underlying problem before machine learning existed, then briefly describe the major developments that eventually led to today's computational approaches. Your historical review should answer, as appropriate:

- What is the earliest known form or root of the problem?
- How did the problem evolve over time?
- Why did it become important?
- When did it become relevant to computing or data analysis?
- What major developments led to the approaches used today?

A useful way to approach this is to work backward from recent papers. Identify the foundational ideas or earlier problems that the modern approaches build upon, and gradually trace their development toward the present. You should prioritize sources that are directly relevant to your project. Textbooks, survey papers, influential papers, and reliable technical sources can all be useful for establishing background.

**Keep it brief:** This is a historical overview, *not* a history thesis. Focus on the major milestones that help the reader understand the evolution of your project topic.

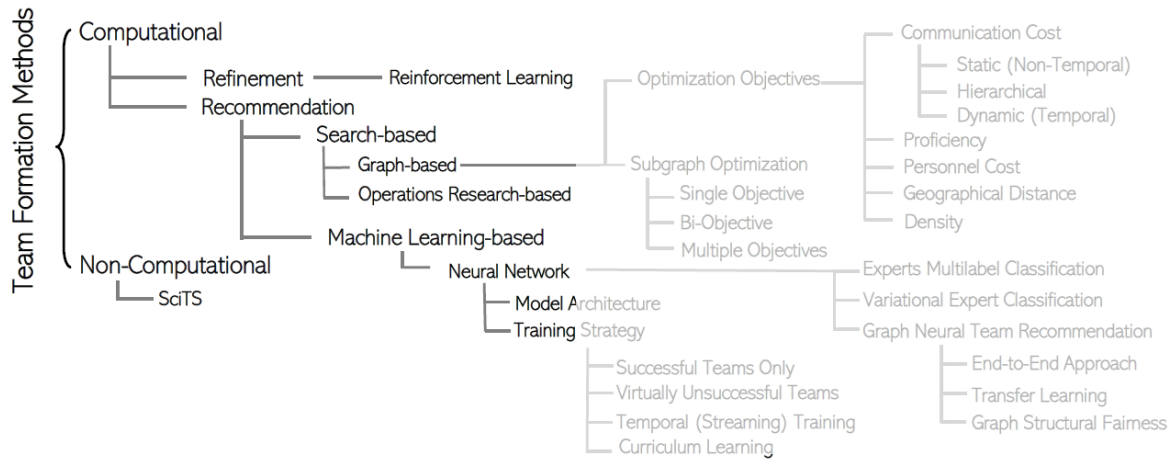


Figure 1. A taxonomy of computational expert team formation methods. For the purpose of this course, the methods shown on the left side are sufficient.

### 1.3. Hierarchy (Categories and Subcategories)

A major step of the literature review is to organize existing approaches into meaningful categories and subcategories. Do *not* simply present existing works or papers in the order in which you found them. Instead, try to identify patterns in how the problem have been approached. You might categorize based on:

- the type of algorithm or model;
- the information or features they use;
- the type of data or domain;
- the learning paradigm;
- the architecture or representation;
- or another perspective that makes sense for your problem.

For example,

[In terms of the type of the algorithm or model]

*“Language modeling can be categorized into two groups: 1) n-gram language models and 2) neural language models.”*

*“Word embedding methods could be categorized into 1) sparse vector representation or 2) dense vector representations.”*

[In terms of the type of domain]

*“Language modeling, based on the underlying genre of communication, would be categorized into 1) formal, 2) scholarly, and 3) informal documents.”*

[In term of information sources]

*“Based on the type of information that employs, language modeling would be categorized into 1) those that use part of speeches (POS), 2) those that use syntactic information, and 3) those that use none.”*

You can choose one perspective at the highest level of the hierarchy. Then, at each branch of the first level, you can continue with the second perspective. Here is an example:

*“Based on the type of document, language modeling methods can be categorized into formal, scholarly, and informal documents. Language models for formal documents*

4

*can be further divided into n-gram language models and neural language models. However, language modelling methods for informal documents are all based on neural methods.”*

Usually, **up to two levels of related works are enough** for a course project. The purpose of the hierarchy is not to create a complicated taxonomy. Its purpose is to help the reader understand the landscape of existing approaches. A hierarchy can be visualized, as shown in Figure 1.

#### 1.4. Comparative Analysis

Once you have established your categories, select representative approaches from each important category and explain them. You do *not* need to summarize every work that you find. Instead, focus on approaches that are:

- influential or widely used;
- representative of a particular category;
- particularly relevant to your project;
- strong baselines that your project may use for comparison; or
- important because they introduced a significant idea.

For each representative approach, briefly explain:

- What does the method do?
- How does it work at a high level?
- What data or information does it use?
- What are its main strengths or advantages?
- What are its important limitations?

You should also explain important developments between approaches. The goal is to explain how the field evolved and why different approaches exist.

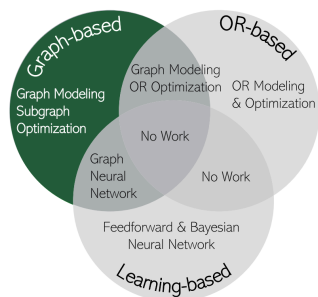
For example:

*“Method A uses hand-crafted lexical features. Method B extends this approach by incorporating syntactic information, which improves performance when lexical information alone is insufficient. Method C replaces hand-crafted features with learned representations, allowing the model to capture ....”*

This is much more useful than writing:

*“Paper A proposed Method A. Paper B proposed Method B. Paper C proposed Method C.”*

A comparison table or a Venn diagram can help a reader for an organized understanding, as shown in Table 1 and Figure 2. However, they are *not* a replacement for the written discussion. Its purpose is to help you identify similarities, differences, strengths, and limitations across approaches.



| Year                | Hybrid | Communication Cost |     |                 |                     |             | Proficiency     |             |                 | Personal Cost | Geographical Distance | Density |
|---------------------|--------|--------------------|-----|-----------------|---------------------|-------------|-----------------|-------------|-----------------|---------------|-----------------------|---------|
|                     |        | Diameter           | MST | Leader Distance | Sum of Edge Weights | Random Walk | Sum of Distance | Trust Score | Expertise Level |               |                       |         |
| Lappas et al. [79]  | 2009   | ✓                  | ✓   |                 |                     |             |                 |             |                 |               |                       |         |
| Li et al. [80]      | 2010   |                    |     | ✓               |                     |             |                 |             |                 |               |                       |         |
| Farhadi et al. [24] | 2011   |                    | ✓   |                 |                     |             |                 |             | ✓               |               |                       |         |
| Kargar et al. [70]  | 2011   |                    |     | ✓               |                     |             | ✓               |             |                 |               |                       |         |
| Datta et al. [17]   | 2012   | ✓                  | ✓   |                 |                     |             |                 |             |                 |               |                       |         |
| Kargar et al. [71]  | 2012   |                    |     |                 |                     |             | ✓               |             |                 |               | ✓                     |         |
| Gajewar et al. [58] | 2012   |                    |     |                 |                     |             |                 |             |                 |               |                       | ✓       |
| Kargar et al. [72]  | 2013   | ✓                  | ✓   |                 |                     |             | ✓               |             |                 |               | ✓                     |         |
| Kargar et al. [1]   | 2013   | ✓                  | ✓   |                 |                     |             | ✓               |             |                 |               | ✓                     |         |

Figure 2. A Venn diagram to highlight overlapping areas where operations research and learning-based methods leverage an expert graph. Gray areas are outside the scope of the current project. Table 1. An overview of objectives employed in subgraph optimizations for expert team formation problem.

## 1.5. Position Your Project

The final step is to explain where your project fits within the landscape of existing approaches. Your goal is not necessarily to claim that your project is completely new. Instead, explain how your project relates to what you have learned from the literature. For example:

*“Existing text-classification approaches can be broadly divided into feature-based and neural methods. Feature-based methods can work well with limited training data but depend heavily on manually designed features. Neural methods can learn representations automatically but may require substantially more training data and computational resources. Our project investigates a neural approach for short, informal social-media text, where limited context presents a particular challenge.”*

Your discussion should make clear:

- Which existing approaches are most relevant to your project?
- Which approaches will you build upon, adapt, compare against, or be inspired by?
- What limitation, opportunity, or practical consideration motivates your project?

If your project is primarily an application-based project, you may not have a research gap. That is perfectly acceptable. Instead, explain what your project contributes. For example, applying an established method to a new domain, or investigating which approach works best for a particular domain.

**Remember:** At this step, you are *not* required to *prove* that your project will advance the state of the art. You are required to demonstrate that you understand the existing work and can clearly explain how your project relates to it.

## 1.6.Submission Guidelines

### 1.6.1. Deliverables

- Submission must be written in English, in the current ACM two-column conference format in LaTeX. Overleaf<sup>3</sup> templates are available from the ACM Website<sup>4</sup>. Use the *sigconf* proceedings template.
- The submission must be *exactly* 2 pages in length, no more and no less, including author list, figures, tables, and references, with unlimited pages for C: AI-Assisted Editing appendix, and must be authored by all team members, with the graduate student listed last in the author list. We understand that fitting

<sup>3</sup> <https://www.overleaf.com/gallery/tagged/acm-official>

<sup>4</sup> <https://www.acm.org/publications/proceedings-template>

all the required content into exactly one page can be challenging. However, this is intentional: an important part of communication is learning how to retouch, refine, condense, and prioritize information without losing its essential meaning. We recommend initially writing a longer version (e.g., 5-10 pages) and then gradually reducing and retouching it until you can effectively communicate the key ideas within the required two-page limit.

- The submission must be a single zip file named: 2\_Background\_firstname1\_firstname2\_[...].zip, e.g., 2\_Background\_Reia\_Ali\_Luiz\_Hossein.zip, containing:
  - o The LaTeX source files and any required supplementary files;
  - o The compiled two-page PDF file.

A *sample* is attached to this manual and is also available [here](#)<sup>5</sup> (2\_Background.tex).

### 1.6.2. Team and Individual Contributions

- Each team must maintain a Git-based repository (e.g., GitLab, GitHub, ...) <sup>6</sup> for the project and linked to the Abstract. The repository must be associated with the individual accounts of all team members. The repository will be used to document the development of the project and to make individual contributions traceable. The repository should be used throughout the project rather than only immediately before a submission deadline. The lab instructor and/or marker may consider the repository's commit history, pull requests, issues, documentation, and other project records when assess individual contributions. The number of commits alone will not be used as a measure of contribution. The quality, significance, and nature of each member's contributions will also be considered.
- The graduate student serves as the project lead, responsible for coordinating the process, maintaining the overall project direction, and ensuring that the project develops toward a meaningful contribution. However, the project lead is also expected to be an active contributor, participating in activities such as problem formulation, literature review, methodology development, implementation, experimentation, analysis, and writing.
- Undergraduate members contribute to the project activities under the guidance of the project lead. All team members are expected to make substantive contributions, and no member should be limited solely to coordination or project management.
- As part of **Milestone 6. Project Management & Documentation (Graduate Student Only)**, the graduate student (project lead) is expected to document each member's contributions throughout the project, with a brief contribution statement provided at each major project milestone. These statements will be compiled into the final project management report.

### 1.6.3. Marking Scheme

- 10% Abstract
- 10% Introduction
- 10% History
- 20% Hierarchy + Figure
- 30% Comparative Analysis + Table + Venn Diagram
- 10% Positioning your project
- 10% Author list, style, format, repo, file naming convention, ...

<sup>5</sup> <https://www.overleaf.com/read/bsrxtwqgggsb>

<sup>6</sup> Web-based applications and cloud services may collect, use and/or disclose your personal information and store or access that information outside of Canada. Personal information includes, but not limited to, identifiable individual's information such as your name or your email address. At UVic, GitLab at [gitlab.uvic.ca](https://gitlab.uvic.ca) is highly recommended for coursework, research and projects. For more information, see <https://www.uvic.ca/systems/services/pages/gitlab.php>. At your team's discretion, you may use GitHub to store information about your project, but we do not require nor recommend storing any personal identifying information. Also, we do not require public repo, but it should be accessible to instructors and markers for evaluation purposes.